



NETWORK DEFENSE WITH OPEN-SOURCE TOOLS

Technical Report & Documentation

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Program: Red Teaming Internship

Week: Week 2



1. Overview

This report documents Activity 5 of Week 2 of the CyArt Red Teaming Internship: Network Defense with Open-Source Tools. The objective was to configure Suricata as a network intrusion detection and prevention system (IDS/IPS), integrate alerts with Elastic SIEM for centralized visibility, and use CrowdSec for collaborative threat intelligence. A custom Suricata rule was written to block a malicious IP, tested from another VM, and the resulting alert was mapped to a MITRE ATT&CK technique.

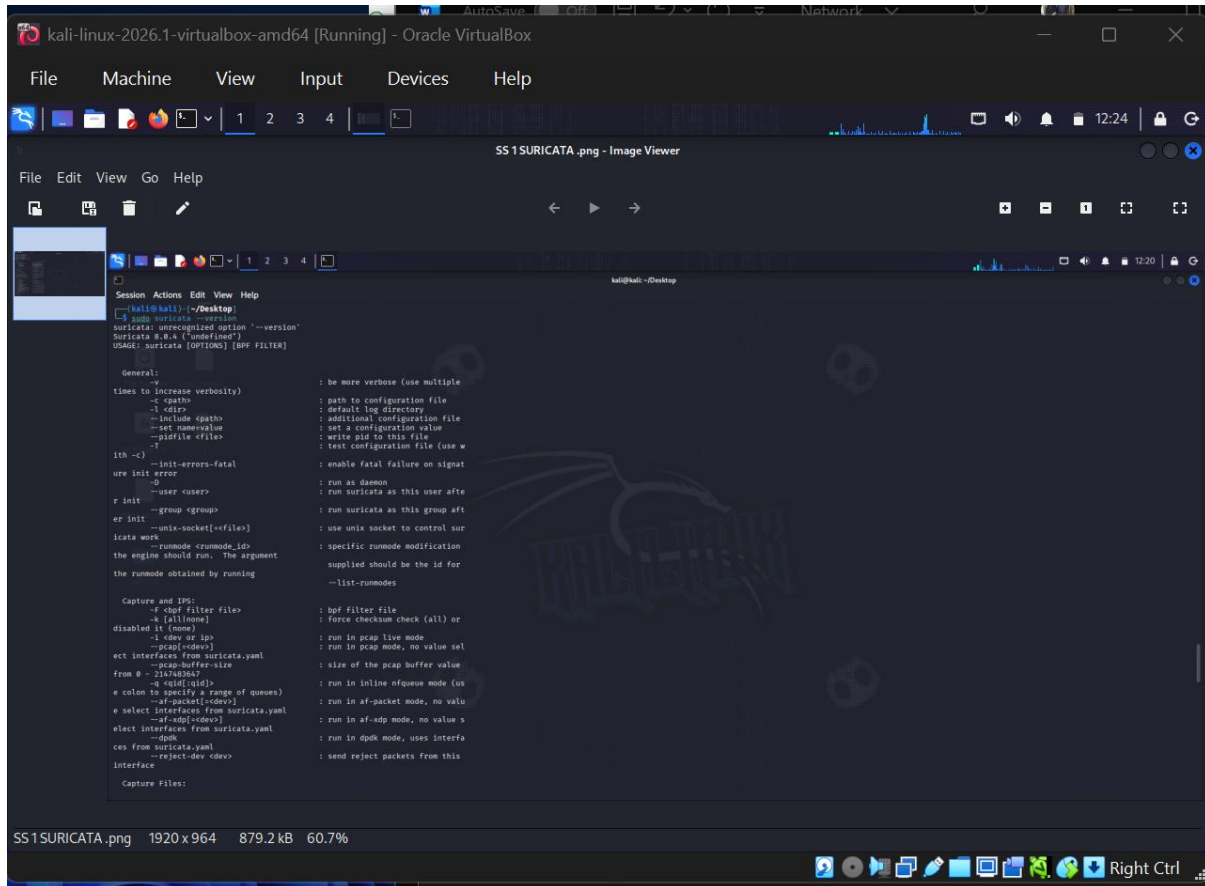
2. Tools Used

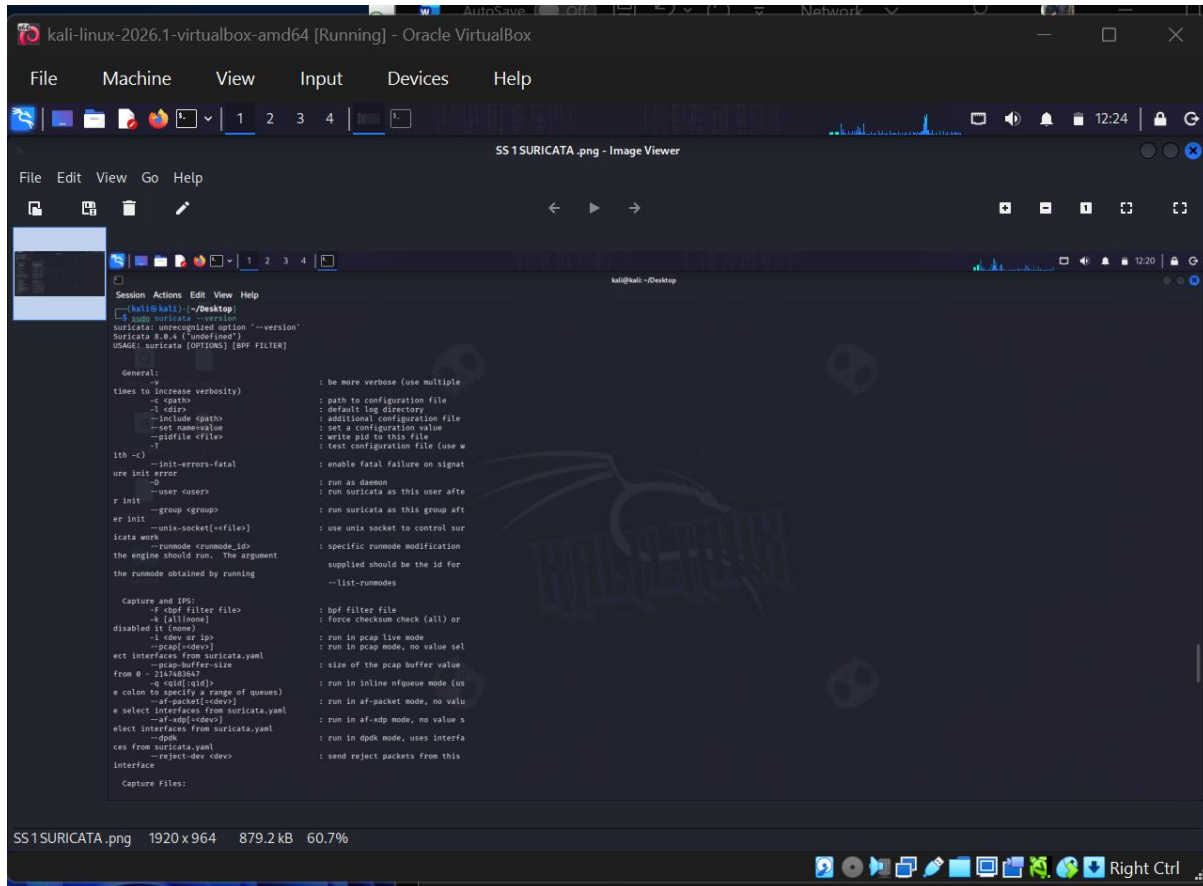
- Suricata — open-source network IDS/IPS used to write and enforce traffic rules
- Elastic SIEM — security information and event management platform for alert ingestion and visualization
- CrowdSec — open-source collaborative IPS with a community threat intelligence feed
- Windows VM / Linux VM — target and attacker machines used for testing the block rule

3. Environment Setup

3.1 Suricata Installation and Configuration

Suricata was installed on the network gateway or monitoring host. The `suricata.yaml` configuration file was edited to set the correct network interface for traffic inspection and to enable the IPS (inline) mode so that the drop rule could actively block packets rather than just alert on them.





3.2 Elastic SIEM Setup

Elastic SIEM was configured to receive Suricata alert logs using the Filebeat Suricata module. The eve.json log file generated by Suricata was ingested into Elasticsearch and made searchable in Kibana for alert analysis and visualization.



The screenshot shows a Kali Linux virtual machine window titled "kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar is a toolbar with icons for file operations. The main window displays the Suricata configuration file located at "/etc/suricata/suricata.yaml" using the GNU nano 8.7.1 editor. The configuration file contains various settings for Suricata, including network interfaces, address groups, and server configurations. The bottom of the window shows a status bar with various keyboard shortcuts and a "Right Ctrl" button.

```
GNU nano 8.7.1 /etc/suricata/suricata.yaml
# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html
# This configuration file was generated by Suricata 8.0.4.
suricata-version: "8.0"

##
## Step 1: Inform Suricata about your network
##

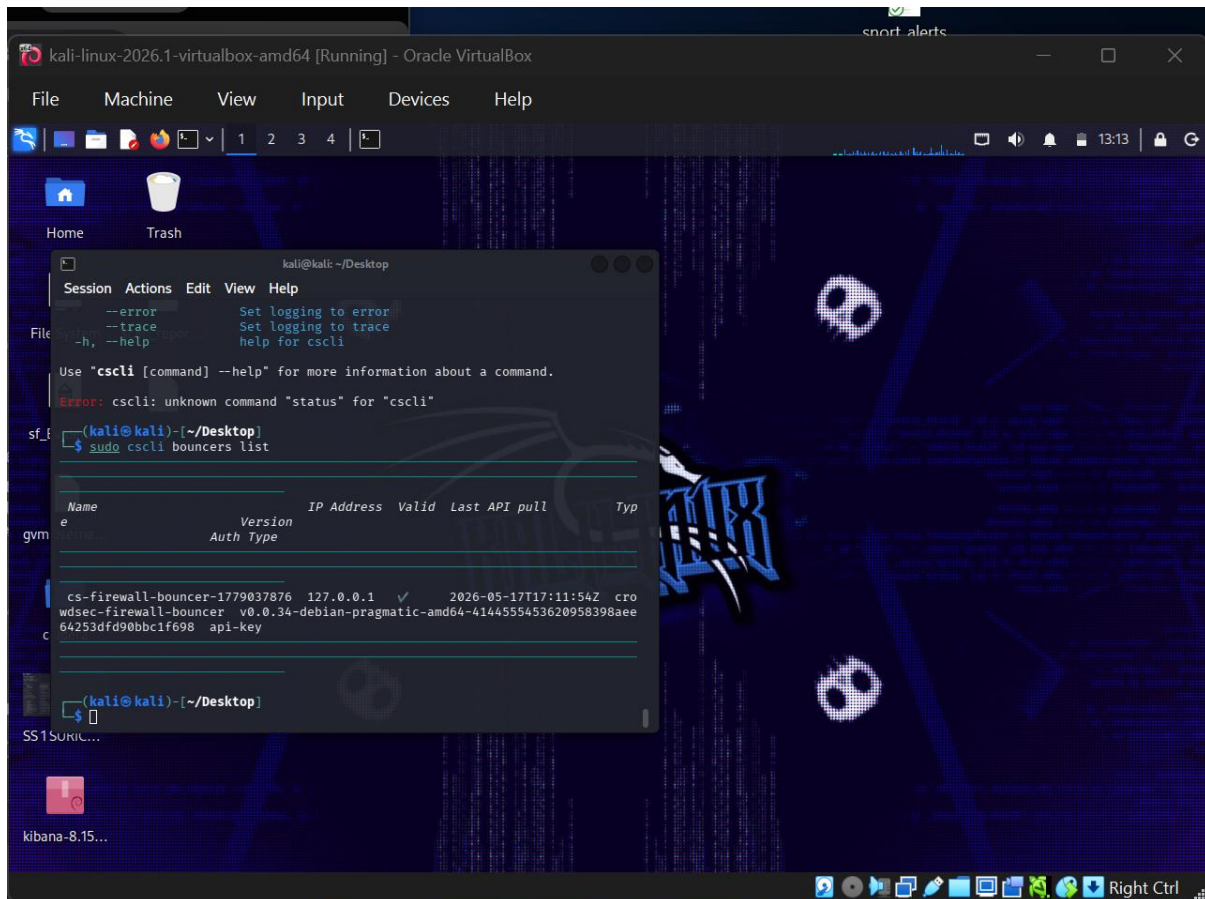
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"

    HTTP_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
    SQL_SERVERS: "$HOME_NET"
    DNS_SERVERS: "$HOME_NET"
    TELNET_SERVERS: "$HOME_NET"
    AIM_SERVERS: "$EXTERNAL_NET"
    DC_SERVERS: "$HOME_NET"
    DNP3_SERVER: "$HOME_NET"
    DNP3_CLIENT: "$HOME_NET"
    MODBUS_CLIENT: "$HOME_NET"
    MODBUS_SERVER: "$HOME_NET"
    ENIP_CLIENT: "$HOME_NET"
    ENIP_SERVER: "$HOME_NET"
```

3.3 CrowdSec Setup

CrowdSec was installed alongside Suricata to provide community-sourced threat intelligence. CrowdSec's blocklist was configured to automatically feed malicious IP ranges into the network defense layer, complementing the custom Suricata rules.



4. Suricata Rule — Blocking a Malicious IP

4.1 Rule Creation

A custom Suricata rule was written to drop (block) all traffic originating from the IP address 192.168.1.100, which was designated as the malicious host in this lab environment. The rule was added to the local rules file at /etc/suricata/rules/local.rules.

Suricata Drop Rule:

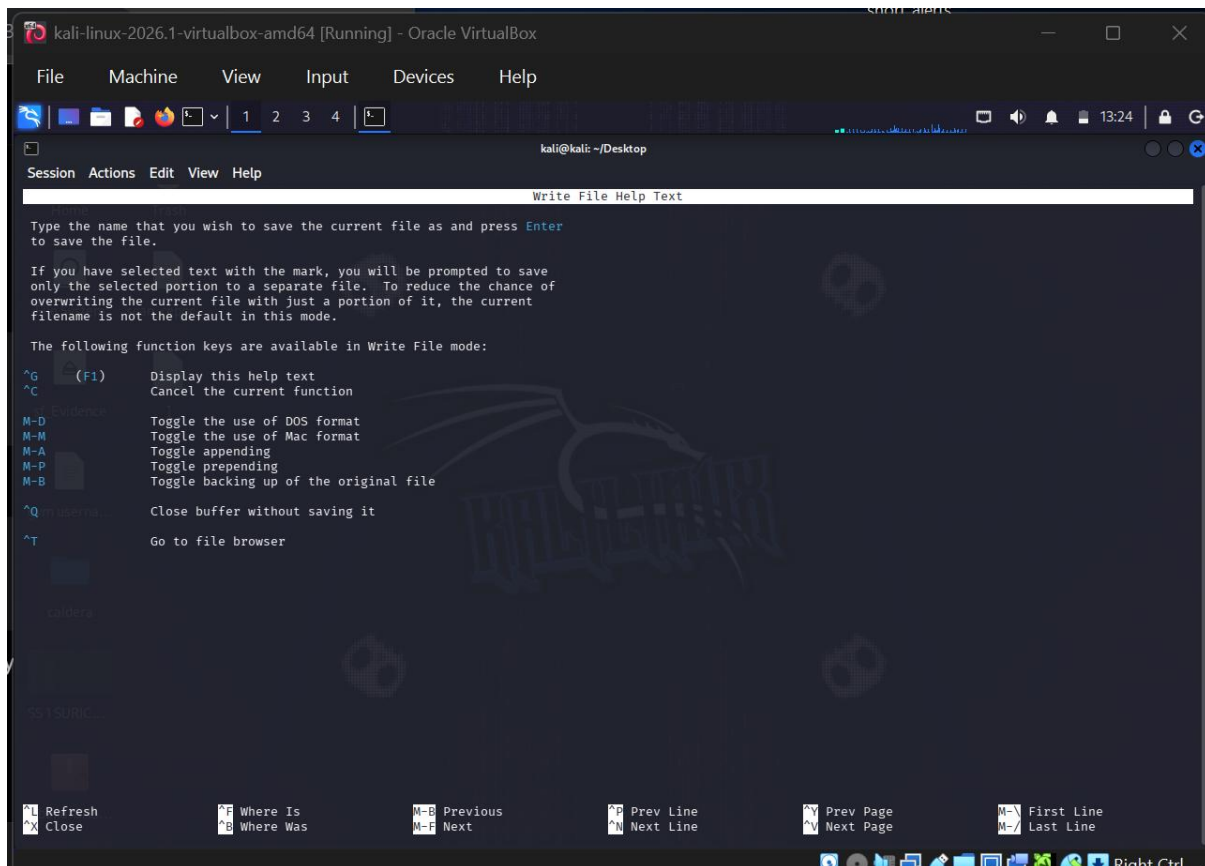
```
drop ip 192.168.1.100 any -> any any (msg:"Block Malicious IP"; sid:1000001;)
```

Rule breakdown:

- drop — action to take: block the packet and prevent it from reaching its destination
- ip 192.168.1.100 any — match any IP traffic from source 192.168.1.100 on any port
- -> any any — heading to any destination IP on any port



- msg:"Block Malicious IP" — human-readable alert message logged when the rule fires
- sid:1000001 — unique Suricata rule identifier (local rules use SIDs starting at 1000001)



4.2 Reloading Suricata with the New Rule

After writing the rule, Suricata was reloaded to apply the changes. The following command was used to reload rules without a full restart:

```
sudo kill -USR2 $(pidof suricata)
```

Alternatively, Suricata was restarted to fully reload the configuration:

```
sudo systemctl restart suricata
```




The screenshot shows a Kali Linux virtual machine desktop. A terminal window is open, displaying the following commands and output:

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Warning: detect: No rule files match the pattern /var/lib/suricata/rules/suricata.rules

(kali@kali)-[~/Desktop]
$ sudo systemctl restart suricata

(kali@kali)-[~/Desktop]
$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; disabled; pre-
   Active: active (running) since Sun 2026-05-17 13:33:55 EDT; 4s ago
   Invocation: 7147484c5824485e9ce460d8c54c8ff1
   Docs: man:suricata(8)
        https://suricata.io/documentation/
  Main PID: 3280 (Suricata-Main)
    Tasks: 8 (limit: 2112)
   Memory: 44.5M (peak: 44.6M)
      CPU: 226ms
   CGroup: /system.slice/suricata.service
           └─3280 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.conf

May 17 13:33:54 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP
May 17 13:33:54 kali suricata[3279]: i: suricata: This is Suricata version 8
May 17 13:33:55 kali systemd[1]: Started suricata.service - Suricata IDS/IDP
lines 1-17/17 (END)
```

4.3 Testing the Block Rule

To verify that the rule was working correctly, a ping was sent from the designated malicious VM (192.168.1.100) to another host on the network. In IPS mode, Suricata dropped the packets and the ping received no replies, confirming the block was active.

```
# Run from VM with IP 192.168.1.100

ping 192.168.1.1
```

Expected result: 100% packet loss, confirming all traffic from the malicious IP was dropped by Suricata at the network level.

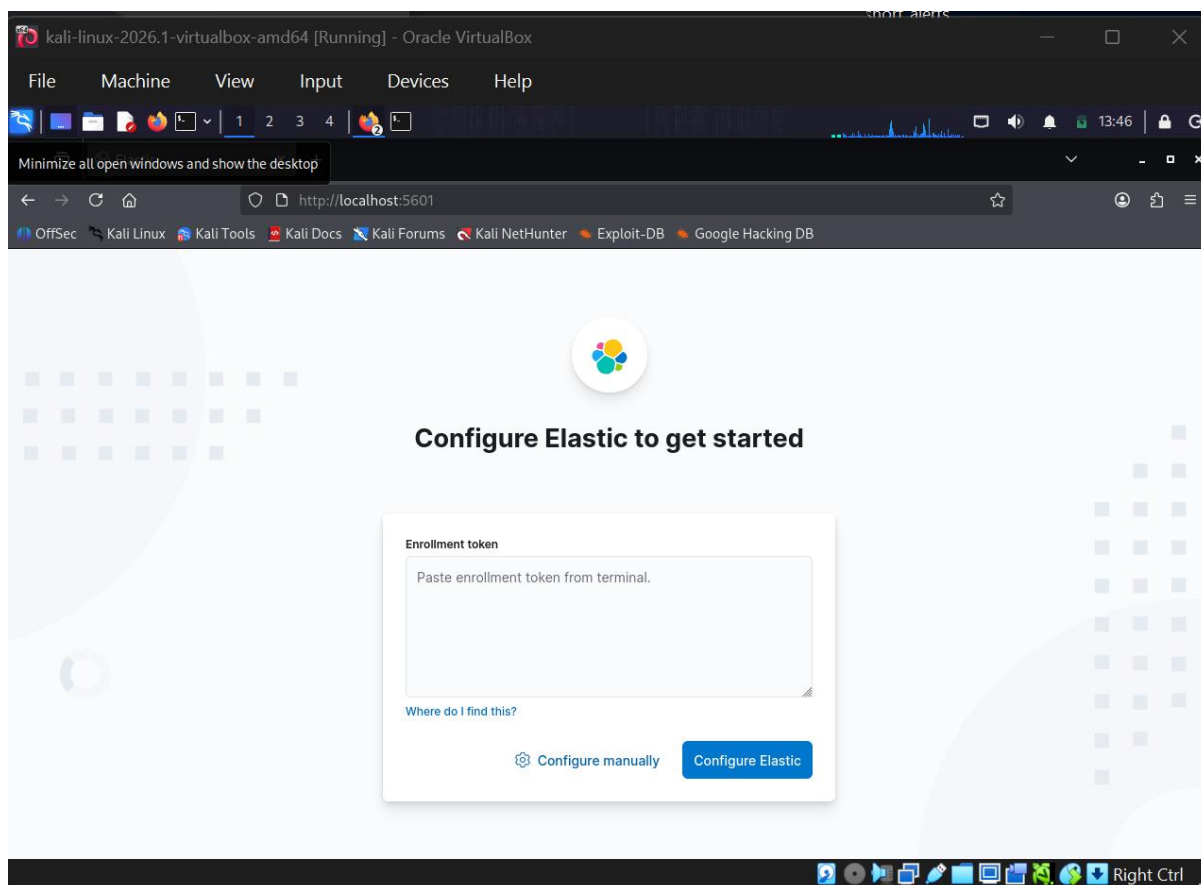


```
kali@kali: ~/Desktop
ping: 192.168.1.105: Name or service not known

(kali@kali)~/Desktop
$ ping 192.168.1.105
PING 192.168.1.105 (192.168.1.105) 56(84) bytes of data.
ping: Warning: invalid tv_usec -2882770779079377576 us
ping: Warning: time of day goes back (-250250194715574 s), taking countermeasures
64 bytes from 192.168.1.105: icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 1 69 8 0 57 d2 61 5a 0 3 8f fc 9 6a 0 0 0 0 d9 96 d 0 0 0 0 0 1
0 11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -2882770779079376715 us
ping: Warning: time of day goes back (-250250194715574 s), taking countermeasures
64 bytes from 192.168.1.105: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 1 69 8 0 57 d2 61 5a 0 3 8f fc 9 6a 0 0 0 0 d9 96 d 0 0 0 0 0 1
0 11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -2882770779079376625 us
ping: Warning: time of day goes back (-250250194715574 s), taking countermeasures
64 bytes from 192.168.1.105: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 1 69 8 0 57 d2 61 5a 0 3 8f fc 9 6a 0 0 0 0 d9 96 d 0 0 0 0 0 1
0 11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -2882770779079375321 us
ping: Warning: time of day goes back (-253548729598899 s), taking countermeasures
64 bytes from 192.168.1.105: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 1 69 8 0 23 33 61 5a 0 6 92 fc 9 6a 0 0 0 0 a 33 e 0 0 0 0 0 10
11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -2882770779079374923 us
ping: Warning: time of day goes back (-253548729598899 s), taking countermeasures
```

5. Alert Verification in Elastic SIEM

After the rule triggered, the Suricata alert was ingested into Elastic SIEM via Filebeat and was visible in Kibana. The alert included the source IP, destination, rule SID, message, and timestamp, providing full visibility into the blocked event.



6. MITRE ATT&CK Mapping

6.1 Mapping Suricata Alerts to ATT&CK Techniques

Each Suricata alert was analyzed and mapped to the relevant MITRE ATT&CK tactic and technique. This mapping helps defenders understand the adversary behavior behind each network alert and prioritize response actions accordingly.

Alert	Tactic	Technique ID	Technique Name	Notes
Suspicious HTTP	Command and Control	T1071	Application Layer Protocol	Outbound traffic to C2 server
Block Malicious IP (sid:1000001)	Initial Access	T1190	Exploit Public-Facing Application	Inbound traffic from known malicious IP blocked
Port Scan Detected	Discovery	T1046	Network Service Discovery	Attacker scanning for open services



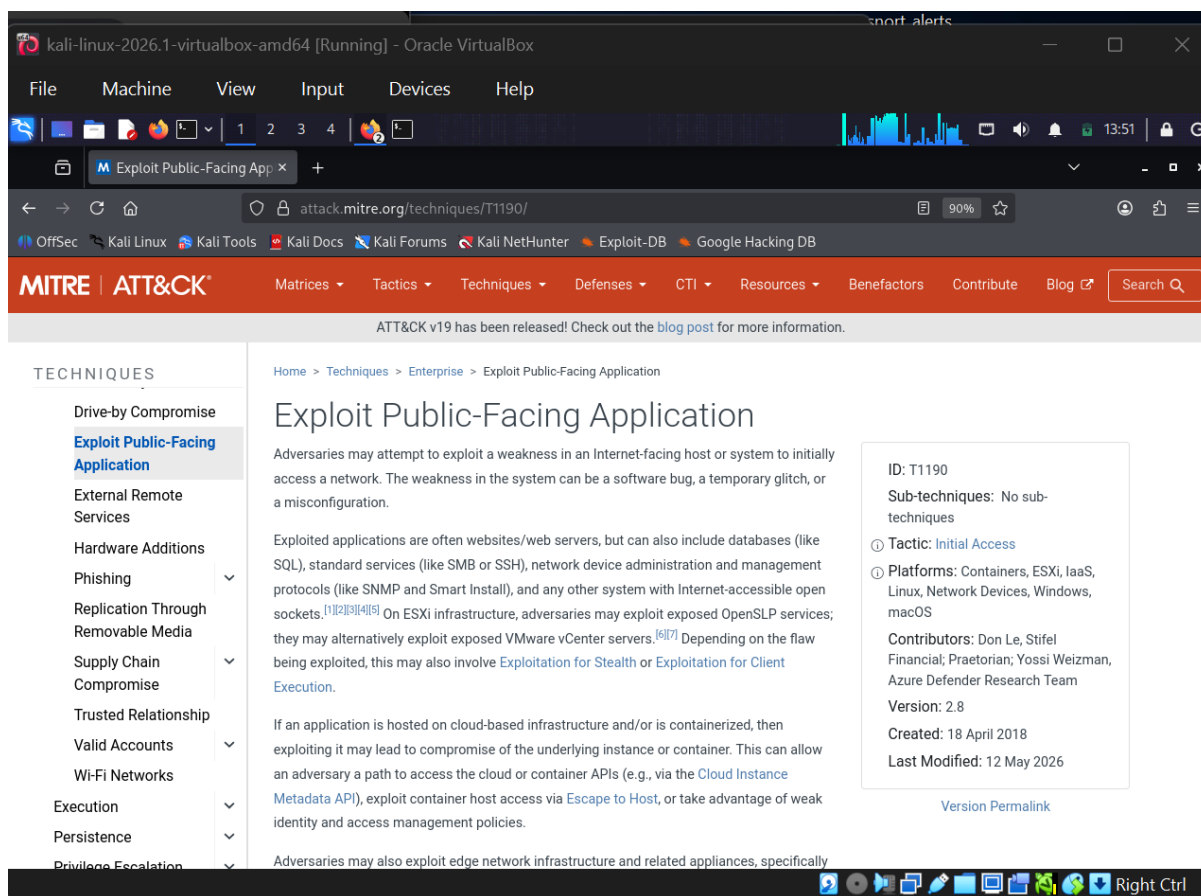
Alert	Tactic	Technique ID	Technique Name	Notes
Suspicious DNS Query	Exfiltration	T1048	Exfiltration Over Alternative Protocol	DNS used as covert exfiltration channel
CrowdSec IP Blocklist Hit	Reconnaissance	T1595	Active Scanning	Community-flagged scanner blocked by CrowdSec bouncer
[Add from your lab]				

Note: Add rows from your actual Suricata and Elastic SIEM alert output during the lab session.

6.2 ATT&CK Technique Reference

The MITRE ATT&CK techniques referenced in the mapping table above are described below for documentation purposes:

ID	Name	Description
T1071	Application Layer Protocol	Adversaries communicate using standard application layer protocols such as HTTP, HTTPS, DNS to blend in with normal traffic.
T1190	Exploit Public-Facing Application	Adversaries exploit weaknesses in internet-facing applications to gain initial access to a network.
T1046	Network Service Discovery	Adversaries scan victim networks to identify open ports and services available on remote hosts.
T1048	Exfiltration Over Alternative Protocol	Adversaries steal data by encoding it within an allowed protocol such as DNS or ICMP to evade detection.
T1595	Active Scanning	Adversaries actively probe victim infrastructure to gather information about systems, ports, and services.



7. CrowdSec Integration Findings

CrowdSec's community blocklist was observed flagging several IPs that were also detected by Suricata rules, providing a second layer of validation. The CrowdSec bouncer enforced blocks at the firewall level, complementing Suricata's inline packet dropping.

- CrowdSec identified community-reported scanner IPs attempting reconnaissance against the lab network.
- The integration between CrowdSec and Suricata demonstrated how layered defenses reduce the attack surface.
- CrowdSec signals were also visible in Elastic SIEM after log forwarding was configured.



```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
[sudo] password for kali:
• kibana.service - Kibana
  Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)
  Active: active (running) since Sun 2026-05-17 13:29:06 EDT; 15min ago
  Invocation: def1e0b6e41b40d3abe903fad9624ffb
  Docs: https://www.elastic.co
  Main PID: 642 (node)
  Tasks: 11 (limit: 2112)
  Memory: 134.5M (peak: 476.8M, swap: 172.6M, swap peak: 172.8M)
  CPU: 15.600s
  CGroup: /system.slice/kibana.service
          └─642 /usr/share/kibana/bin/node /usr/share/kibana/bin/.. /src/cli/dist

May 17 13:29:12 kali kibana[642]: Native global console methods have been overridden in production environment.
May 17 13:29:15 kali kibana[642]: [2026-05-17T13:29:15.476-04:00][INFO ][root] Kibana is starting
May 17 13:29:15 kali kibana[642]: [2026-05-17T13:29:15.504-04:00][INFO ][node] Kibana process configured with roles: [background_tasks, ui]
May 17 13:29:28 kali kibana[642]: [2026-05-17T13:29:28.077-04:00][INFO ][plugins-service] The following plugins are disabled: "cloudChat,cloudExperiments,
May 17 13:29:28 kali kibana[642]: [2026-05-17T13:29:28.148-04:00][INFO ][http.server.Preboot] http server running at http://localhost:5601
May 17 13:29:28 kali kibana[642]: [2026-05-17T13:29:28.302-04:00][INFO ][plugins-system.preboot] Setting up [1] plugins: [interactiveSetup]
May 17 13:29:28 kali kibana[642]: [2026-05-17T13:29:28.325-04:00][INFO ][preboot] "interactiveSetup" plugin is holding setup: Validating Elasticsearch con
May 17 13:29:28 kali kibana[642]: [2026-05-17T13:29:28.361-04:00][INFO ][root] Holding setup until preboot stage is completed.
May 17 13:29:37 kali kibana[642]: i Kibana has not been configured.
May 17 13:29:37 kali kibana[642]: Go to http://localhost:5601/?code=561322 to get started.

(kali@kali)-[~]
$ sudo cscli decisions list
No active decisions

(kali@kali)-[~]
$ sudo cscli decisions add --ip 192.168.1.100 --reason "FTP Attack"
INFO Decision successfully added

(kali@kali)-[~]
$ sudo cscli decisions list



| ID | Source | Scope:Value      | Reason     | Action | Country | AS | Events | expiration | Alert ID |
|----|--------|------------------|------------|--------|---------|----|--------|------------|----------|
| 1  | cscli  | Ip:192.168.1.100 | FTP Attack | ban    |         |    | 1      | 3h59m52s   | 1        |



(kali@kali)-[~]
$
```

8. Key Observations and Findings

- Suricata's drop rules work effectively in IPS mode, actively blocking traffic rather than just logging it as in IDS mode.
- Custom local rules give defenders fine-grained control over which traffic to block based on IP, port, protocol, or payload signatures.
- Elastic SIEM provided a centralized and searchable view of all Suricata alerts, making it easy to identify patterns and investigate incidents.
- CrowdSec added a community-driven intelligence layer that blocked known malicious actors without requiring manual rule writing.
- Mapping alerts to MITRE ATT&CK transforms raw network events into actionable threat intelligence by linking them to known adversary behaviors.

9. Challenges Faced

- Configuring Suricata in IPS mode required setting up NFQ (NetFilter Queue) correctly on Linux, which needed kernel module configuration.
- Ensuring Filebeat correctly parsed Suricata's eve.json format required verifying the Filebeat Suricata module configuration.



- Mapping some Suricata alerts to ATT&CK techniques required understanding both the rule intent and the broader adversary context.

10. Learnings and Takeaways

- Network defense tools like Suricata operate at the packet level and can both detect and prevent malicious traffic in real time.
- Writing custom Suricata rules requires understanding of network protocols, IP addressing, and the Suricata rule syntax.
- Elastic SIEM transforms raw logs into searchable, visual intelligence that dramatically speeds up incident investigation.
- CrowdSec demonstrates how community-sourced threat intelligence can augment individual organization defenses.
- The MITRE ATT&CK framework is an essential tool for contextualizing alerts and communicating findings to both technical and non-technical stakeholders.

11. Next Steps

- Write additional Suricata rules for common attack signatures such as SQL injection, XSS, and directory traversal patterns.
- Build a Kibana dashboard in Elastic SIEM to visualize Suricata alerts by tactic, technique, and source IP over time.
- Explore Suricata's file extraction feature to capture and analyze suspicious files transferred over the network.
- Integrate CrowdSec metrics into the Elastic SIEM dashboard for a unified defense overview.